

EL1: Intro

Overwiev of this part of the course

Important aspects

- Data (where does it come from, what does it contain)
- Ethics and legal (how to handle sensitive data, what laws and regulations apply)
- Project management (how to plan and execute a data project, version control, reproducibility, R specific packages for efficient data handling)

Data – what is it?

EU Data Act | Article 2, Definitions:

For the purposes of *this Regulation*, the following definitions apply:

- (1) ‘data’ means any digital representation of acts, facts or information and any compilation of such acts, facts or information, including in the form of sound, visual or audio-visual recording;
- (2) ‘metadata’ means a structured description of the contents or the use of data facilitating the discovery or use of that data;
- (3) ‘personal data’ means personal data as defined in Article 4, point (1), of Regulation (EU) 2016/679;
- (4) ‘non-personal data’ means data other than personal data;

Course structure

- Lectures on different data sources/registers
- 🧑‍🔬 Exercises on data management and analysis
 - R with some additional tools (Git, GitHub, targets, data.table)
- A data project with written report and presentation
- Final exam 🧑‍🎓
- Instruction web page [in addition to Canvas](#)
- Litterature: accessible through [GU library](#) ([O’Reilly Learning for Higher Education](#)) or otherwise shared (no need to purchase books)

Different types of data

-  Images
 - Statistical image analysis
-  Lab samples
-  Unstructured medical records
 - Natural Language Processing
-  Sensor data
 - Time series (“big data”)
-  EHRs (electronic health records)
 - Structured but hierarchical rather than tabular
-  **Structured medical records**
 - tabular data

Usages

-  Research
-  Quality control/improvement
-  Administration/reporting
-  News coverage
-  Building prediction models and tools

Register data

Three types of health care registers:

- Administrative registers
- Health care registers
- Quality registers

Administrative data

(As found in all types of registers)

- Billing codes
 - Direct (what something actually cost)
 - Estimated (DRG codes for different types of procedures)
- Claims data
 - Primary for reimbursement (insurance company or other payer)
 - Secondarily for Health economy/epidemiology
- How to contact patients, health care providers etc
- Dates and times for visits, procedures etc



Hospital background data

- hospital characteristics
- staffing
- resources
- geographical area
- level of specialization
- private, public



Clinical data

- health care registers
 - Mandatory (by law)
 - eg: National patient register, cancer register (diagnoses)
- quality registers
 - Optional for health care providers
 - (Mandatory within organizations joining)
 - conditions (diabetes, cancer, etc)
 - procedures (total hip arthroplasty)
 - Diagnoses, treatments, health status, questionnaires (PROM/PREM)



Individual background data

- socioeconomic data
- education
- income
- occupation
- family relations
- migration status
- Mortality data
 - date of death
 - cause of death



Aggregated data

“Micro” vs. “macro” data.

- population data
- neighborhood characteristics
- pollution
- crime rates

Inclusion/exclusion criteria

- 👍 Defines the target study/register population

- 🙋 Define exceptions to the general rules

🔪 Simple example

“Every Swedish resident who had total hip arthroplasty performed in Sweden”

- **Include:** all ages, all hospitals, all reasons for the prosthesis, all types of prosthesis
- **Exclude:** Swedish residents with surgery performed in other countries. Non-Swedish residents with the procedure performed in Sweden.

😓 Complicated example

[The National Quality Register for Ovarian Cancer](#)

• Inclusion

1. Epithelial borderline tumours of the ovary

- Topography code according to ICD-O/2: C56.9.
- Morphology code according to ICD-O/2 ≥ 80103 and < 85900 .
- Borderline tumours with 5th digit 3 in the morphology code according to ICD-O/2 and benign behaviour flag = 3.

2. Epithelial ovarian cancer:

- Topography code according to ICD-O/2: C56.9.
- Morphology code according to ICD-O/2 ≥ 80103 and < 85900 .
- Malignant tumours with 5th digit 3 in the morphology code according to ICD-O/2 and benign behaviour flag blank.

3. Non-epithelial ovarian cancer:

- Topography code according to ICD-O/2: C56.9.
- Morphology code according to ICD-O/2 ≥ 85903 and < 95900 , with the exception of mesotheliomas with ICD-O/2 codes in the interval ≥ 90500 and < 90600 .
- Malignant tumours with digit 3 as the fifth digit in the morphology code according to ICD-O/2.
- Exception for granulosa cell tumours, where all cases with morphology codes according to ICD-O/2 in the interval ≥ 86200 and ≤ 86223 are included.

4. Malignant tumours of the fallopian tube:

- Topography code according to ICD-O/2: C57.0.
- Morphology code according to ICD-O/2 ≥ 80003 and < 95900 , with the exception of mesotheliomas with ICD-O/2 codes in the interval ≥ 90500 and < 90600 .
- Malignant tumours with digit 3 as the fifth digit in the morphology code according to ICD-O/2.

• Exclusion

▸ Epithelial ovarian cancer and borderline tumours of the ovary

Cases with behavior codes **0, 1, 2, 6, or 9** as the fifth digit in the ICD-O/2 morphology code

are excluded.

Morphology codes according to ICD-O/2 <80103 and ≥85900 are excluded.

▸ **Non-epithelial ovarian cancer**

Cases with **digits 0, 1, 2, 6, or 9 as the fifth digit** in the ICD-O/2 morphology code are excluded, **with the exception of granulosa cell tumours**, for which cases with ICD-O/2 morphology codes in the interval ≥86200 and ≤86223 are included even when the final digit is **0, 1, 2, or 3**.

Morphology codes according to ICD-O/2 <85903, as well as codes in the intervals ≥90500 and <90600 (mesotheliomas) and ≥95900, are excluded.

▸ **Tumours of the fallopian tube**

Cases with **behaviour codes 0, 1, 2, 6, or 9 as the fifth digit** in the ICD-O/2 morphology code are excluded.

Morphology codes according to ICD-O/2 in the intervals ≥90500 and <90600 (mesotheliomas) and ≥95900 are excluded.

▸ **For all diagnoses**, cases are excluded if the diagnosis is based solely on:

- clinical examination (**basis of diagnosis 1**),
- imaging procedures including radiography, scintigraphy, ultrasound, MRI, CT (or equivalent examinations) (**basis of diagnosis 2**),
- autopsy with or without histopathological examination (**basis of diagnosis 4 or 7**),
- surgery without histopathological examination (**basis of diagnosis 6**), or
- other laboratory investigations (**basis of diagnosis 8**).
- cases with **age <18 years** are excluded.

Coverage and completeness

- 🏥 **Institutional coverage**: proportion of all eligible units/clinics that are connected to the registry
 - e.g., 90% of hospitals performing the procedure are connected
 - Should be known by the “register holder”
- 😞 **Case coverage**: proportion of patients who should have been reported from connected units that are actually included
 - e.g., 85% of eligible patients registered
 - The aim is to use 100 % but this is not always possible
- **Data completeness**: proportion of required data fields that are filled in for the registered patients
 - 🚬 e.g., 95% of patients have smoking status recorded
 - 🩸 e.g., 80% of patients have blood pressure data available

What is recorded?

- 👤 Some registers are mandated by law and regulations
- Quality registers often have a steering committee and register holder
- Research initiated databases according to specific protocols

Data linking

- Unique personal identifier
 - Not in every country!
 - Social security number similar purpose but not as widely used
- study specific id number
- HSA (“Hälsö- och sjukvårdens adressregister” for staff and organizations)

Unique personal identifier

(Swedish: personnummer, reading: [1])

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- 10 (or 12) digits
- date of birth-4 digits
- assigned at birth or immigration
- used in all health care contacts
- used for all administrative data
- sometimes reused after death
- sometimes changed (uncommon)
- sometimes inclusion criteria for register
- similar in the Nordic countries
 - Denmark: CPR number
 - Norway: Fødselsnummer
 - Finland: Henkilötunnus
 - Iceland: Kennitala

Combining data

- Similar registries in different areas/regions/countries
 - Different individuals but similar data
- Same definitions and variables?
- Same inclusion criteria?
- Don't get fooled by similar names!
- Differences and similarities within the Nordic countries [2]

Working with health care data

A lot to do before the statistical analysis!

- **Legalities**
 - Do I have the right to access this data?
 - What am I allowed to do?
 - What am I not allowed to do?
- **Data management**

- large datasets
- multiple datasets
- different formats
- missing data
- data cleaning
- data transformation
- data wrangling
- data munging
- data governance
- data engineering
- **Planning**
 - What is the purpose?
 - How can I achieve my goals?
 - What if I change my plans later?
 - Can I redo my analysis?
 - How do I present/communicate my results?

R as a tool but ...

- Large files often comes exported from SAS (initially “Statistical Analysis System”)
- Comma-Separated Values (csv) or text files
- Application Programming Interface (API) calls
- Structured Query Language (SQL) databases
- Hierarchical data structures (eXtensible Markup Language, XML; JavaScript Object Notation, JSON, ...)

Our use of R

- `{data.table}` to handle large data sets efficiently
 - `{targets}` to streamline a reproducible pipeline
 - Git for version control
 - GitHub for collaboration
 - Quarto for reporting
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Bibliography

- [1] J. F. Ludvigsson, P. Otterblad-Olausson, B. U. Pettersson, and A. Ekbom, “The Swedish personal identity number: Possibilities and pitfalls in healthcare and medical research,” *European Journal of Epidemiology*, vol. 24, no. 11, pp. 659–667, 2009, doi: [10.1007/s10654-009-9350-y](https://doi.org/10.1007/s10654-009-9350-y).
- [2] K. Laugesen *et al.*, “Nordic Health Registry-Based Research: A Review of Health Care Systems and Key Registries,” *Clinical Epidemiology*, pp. 533–554, Jul. 2021, doi: [10.2147/CLEP.S314959](https://doi.org/10.2147/CLEP.S314959).